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Local Interpretations of Epistemic Logic for Multi-Agent Games

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Abstract

Multi-agent games can be used to model a wide range of problems, from distributed computing to model checking. Reasoning on such games require to look at agents' knowledge and its evolution throughout the game. A formal approach to do so is it to rely on propositional epistemic logic. To evaluate propositional formulas, a model of agents' knowledge is then required. A simple method is to rely on perfect recall, and each agent maintains a complete history of its previous actions and observations.

In this thesis, we aim to introduce more lightweight models that allow local evaluations of epistemic formulas without relying on full histories. We introduce three models derived from the multi-agent knowledge-based subset construction. Two of them are equivalent and based on local knowledge structures that can be maintained throughout the game. They can be used to evaluate formula of epistemic logic with bounded number of knowledge operators. The other one relies on game expansion through multi-agent knowledge-based subset construction, and we conjecture it can also be used to evaluate any formula with bounded number of knowledge operators.

By design, these models can be more lightweight than history-based approach for games running for large number of steps.

Keywords

Multi-agent games, Epistemic logic, Modal logic, Multi-agent systems, Game theory

Sammanfattning

Multiagentspel kan användas för att modellera ett brett spektrum av problem, från distribuerad databehandling till modellkontroll. För att resonera kring sådana spel måste man titta på agenternas kunskap och hur den utvecklas under spelets gång. En formell metod för att göra detta är att förlita sig på propositionell epistemisk logik. För att utvärdera propositionsformler krävs en modell av agenternas kunskap. En enkel metod är att förlita sig på perfekt återkallelse, och varje agent upprätthåller en komplett historik över sina tidigare handlingar och observationer.

I den här avhandlingen försöker vi introducera mer lättviktiga modeller som möjliggör lokala utvärderingar av epistemiska formler utan att förlita sig på fullständig historik. Vi introducerar tre modeller som härrör från den kunskapsbaserade delmängdskonstruktionen för flera agenter. Två av dem är ekvivalenta och baserade på lokala kunskapsstrukturer som kan bibehållas genom hela spelet. De kan användas för att utvärdera formeln för epistemisk logik med begränsat antal kunskapsoperatorer. Den andra förlitar sig på spelexpansion genom multiagentkunskapsbaserad delmängdskonstruktion, och vi antar att den också kan användas för att utvärdera alla formler med ett begränsat antal kunskapsoperatorer.

Genom sin utformning kan dessa modeller vara mer lättviktiga än historikbaserade metoder för spel som löper över ett stort antal steg.

Nyckelord

Multiagentspel, Epistemisk logik, Modal logik, Multiagentsystem, Spelteori

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List of acronyms and abbreviations

MAGIIAN Multi-Agent Games with Imperfect Information Against Nature

Chapter 1

Introduction

This master thesis is about Multi-Agent Games. That is to say games in which a team of entities called agents needs to collaborate against an adversary to complete a common objective. To do so, a strategy can be established by the agents before the game starts. At each step of the game, agents make decisions based on their current knowledge by referring to their strategy, and then can update their knowledge based on the new situation.

More precisely, we are interested in Multi-Agent Games with Imperfect Information Against Nature (MAGIIAN) [1], that are Multi-Agent Games where each agent might have different and imperfect information about the state of the game. Strategies for such games can be represented as a set of Epistemic Logic formulas mapped to actions. For instance, and after converting it to a knowledge-based program:

```
if knowledge respects condition_1 then
    do action_1
else if knowledge respects condition_2 then
    do action_2
...
```

However, one might want to introduce higher-order knowledge. For instance, the 2nd order knowledge for agent 1 in a game with 2 agents would be: agent 1 knows something about the game state AND knows that agent 2 knows something else. Iterating one more time would give 3rd order knowledge, then 4th, 5th, etc. In Multi-Agent Games with Imperfect Information, higher-order knowledge appears to be often required to find a winning strategy since an agent do not have perfect information of the game.

1.1 Problem

The standard representation of knowledge for epistemic logic is a history-based structure relying on a global Kripke structure, common to each agent. However, such a structure cannot practically be used for higher-order knowledge representation as it explodes in terms of complexity, both space-wise and time-wise. Another approach used for epistemic logic is the use of k-trees [2], which have the advantage of being a smaller structure than history-based structures, as the size is independent from the number of actions taken during the play. However, k-trees can only be used for bounded knowledge order with bound k. Hence the need to explore other structures for practical uses.

1.2 Objective

This project aims to prove that an iteratively computed local knowledge representation that is more compact than traditional approaches is sufficient to evaluate epistemic logic formulas. Such a structure would allow for more use cases as the complexity would be reduced. More precisely, the project will rely on an unpublished manuscript by Gurov et al which presents multiple representations [3]. In this report, three representations will be discussed and compared, and the expressive power of some of them will be proven.

1.3 Research Methodology

Due to the nature of the problem, the approach will rely on mathematics and formal proofs. Elements from game theory, formal language and propositional calculus will be used, as they are used to mathematically formalise MAGIIAN and *epistemic logic*. More details and practical information can be found in chapter 3.

1.4 Delimitations

The game studied in this work are limited to MAGIIAN as they are defined in [1]. Among the assumptions made and explicitly stated in chapter 2, the main ones are the following: the set of states of each game is considered finite, and agents cannot communicate. The knowledge representations and interpretations studied in this report are limited to the one defined by Gurov

et al. unpublished manuscript [3]. They are derived from the knowledge structures, which results from the Multi-agent Knowledge Based Subset Construction described in [1]. In this report, no evaluation of the complexity of the different models studied will be made. Only their limitations, and their soundness and completeness w.r.t *epistemic logic* will be studied.

1.5 Structure of the thesis

Chapter 2 presents background information about *epistemic logic* and MAGIIAN, with all the . Chapter 3 details the main mathematical tools used in this report. Chapter 4 presents the proofs and discussions concerning the different interpretations of epistemic logic (or sub logics) for MAGIIAN.

1.6 General Notations

In this report, sequences of equivalent propositions will be used, be it in examples or in proofs. Such sequences will have the following format:

$$
\begin{aligned}
& Proposition_A \\
\iff\ & Proposition_B && \textit{(Argument)} \\
\iff\ & ...
\end{aligned}
$$

Where *Argument* is a succinct argument to explain the corresponding step of the sequence of equivalences.
In addition, standard notations from logics will be used, such as $\vee$ for a logical disjunction, or $\wedge$ for a logical conjunction.
For any set S, the power set will be noted 2^S and for any element $s \in S$, $\downarrow s$ denotes the lower closure (or downward closure).
For any tuple t and any natural n, $t[n]$ denotes the n-th coordinate of t. Coordinates are numbered starting from 1.

Chapter 2

Background

This chapter provides basic background information about epistemic logic in section 2.1. Additionally, it provides an introduction to Multi-Agent Games with Imperfect Information Against Nature (MAGIIAN). in section 2.2. Related works are then discussed in section 2.5.

2.1 Epistemic Logic

This section provides definitions for *epistemic logic* and how formulas are interpreted. More precisely, it provides the syntax of the logic, and its semantics to evaluate formula using Kripke structures.

2.1.1 Syntax and Sub-Logics

Definition 2.1.1 (Epistemic Logic)**.**
Let Var be a set of variables, $\mathcal{AP}$ a set of atomic propositions and $Agt = \{1, 2, \ldots, n\}$ a team of agents. The syntax of epistemic logic is:

$$\phi ::= \ p \mid \neg\phi \mid \phi \vee \phi \mid K_i\phi$$

where the K_i symbol is called a *knowledge operator*, and $K_i\phi$ is read "*agent i knows that ϕ holds*".

Example 1. Example of a possible derivation sequence for an epistemic logic formula:

$$\phi \rightarrow \phi \vee \phi \rightarrow p_1 \vee \phi \rightarrow p_1 \vee K_2\phi \rightarrow p_1 \vee K_2 p_2$$

Hence $p_1 \vee K_2 p_2$ is a formula that can be expressed in epistemic logic (assuming $p_1, p_2 \in \mathcal{AP}$ and at least two agents for K_2 to be defined). This formula can be read as "p_1 *holds or agent* 2 *knows that* p_2 *holds*".

Let $\mathcal{L}_n$ denote the logic defined by the previous syntax. $\mathcal{L}_n$ is the *epistemic logic* we are interested in. Then for any formula in $\mathcal{L}_n$, we define the *alternation depth* as the maximum number of alternation between distinct knowledge operators in the formula plus one. For instance, $K_1 K_2 K_1 p$ has an alternation depth of 3 and $K_2 K_2 K_1 K_2 p$ too, since the first two knowledge operators are identical. $\mathcal{L}_n^{jK}$ denotes the sub-logic of $\mathcal{L}_n$ with formulas with alternation depth at most j. Although the final objective would be to evaluate formulas from *epistemic logic*, this sub-logic will be used for intermediate results.

Note that if they are not directly present in the syntax, binary operators corresponding to the other logic gates can be derived from the unary operator $\neg$ (not) and the binary operator $\vee$ (or).

Now that the syntax is defined, we must precise how the formulas should be evaluated.

2.1.2 Kripke Structures

The formulas are interpreted over Kripke structures, which are defined by the following generic definition:

Definition 2.1.2 (Kripke Structure)**.**
Let $\mathcal{AP}$ be a set of atomic proposition, a Kripke structure is a tuple $\mathcal{M} = (S, \{\rightarrow_i\}_{i \in Agt}, L)$ where:

- S is a set of states, or possible worlds.
- $Agt = \{1, 2, \ldots, n\}$ is a set of agents.
- For each $i \in Agt$, $\rightarrow_i$ is a binary accessibility relation. The precise definition and properties of this accessibility relation will vary from one Kripke structure to another. When the accessibility relation is an equivalence relation, the symbol used will be $\sim_i$.
- $L : S \longrightarrow 2^{\mathcal{AP}}$, where $2^{\mathcal{AP}}$ is the power set of $\mathcal{AP}$, is a labelling function that assigns to each state the set of atomic propositions that are true in it.

This definition of a Kripke structure accounts for the multi-agent setting of this report by the mean of the accessibility relation defined as a set of local accessibility relations. The exact definition of these local accessibility relations will change from one model to another, but they will always be used to evaluate knowledge operators.

Now that both the syntax and the Kripke structures have been defined, we can define the semantics for epistemic logic.

2.1.3 Semantics

Definition 2.1.3 (Epistemic Logic Semantics)**.**
Let $\mathcal{M} = (S, \{\rightarrow_i\}_{i \in Agt}, L)$ be a Kripke structure. The semantics of epistemic logic is defined by induction as:

$$
\begin{aligned}
[\![p]\!]^{\mathcal{M}} &= \{s \in S \mid p \in L(s)\} \\
[\![\neg\phi]\!]^{\mathcal{M}} &= S \setminus [\![\phi]\!]^{\mathcal{M}} \\
[\![\phi \vee \psi]\!]^{\mathcal{M}} &= [\![\phi]\!]^{\mathcal{M}} \cup [\![\psi]\!]^{\mathcal{M}} \\
[\![K_i\phi]\!]^{\mathcal{M}} &= \{s \in S \mid \forall s' \in S, (s \rightarrow_i s' \implies s' \in [\![\phi]\!]^{\mathcal{M}})\}
\end{aligned}
$$

The notation $[\![\, . \,]\!]$ is used to denote semantics, to make a clear distinction between a formula and its semantical meaning, the latter being the set of states in which the formula is true.

Intuitively, *p* is true at a state s if *p* is in the labels of s (refer to Def. 2.1.2 of Kripke structures and labelling function). The negation operator $\neg$ and the logical disjunction operator ('or' operator) $\vee$ are directly interpreted using the equivalent operators in set theory. The K_i operators are the ones used to express knowledge. In English, each operator K_i reads *agent i knows*. A sequence of consecutive K_i operators is read starting from the leftmost operator. For instance, K_1K_2p reads as *agent 1 knows that agent 2 knows that p holds*. Consecutive identical K_i operators can be collapsed to a single K_i operator. For example, $K_2K_2K_1K_2p$ can be reduced to $K_2K_1K_2p$. Intuitively, $K_i\phi$ is true if ϕ is true for any world that agent i considers possible.

To define whether a formula is true in a specific state, we define a satisfaction relation.

Definition 2.1.4 (Satisfaction Relation)**.**

Let $\mathcal{M} = (S, \{\rightarrow_i\}_{i \in Agt}, L)$ be a Kripke structure, and $s \in S$. For any formula $\phi \in \mathcal{L}_n$ the *satisfaction relation* $\mathcal{M}, s \models \phi$ is defined by:

$$\mathcal{M}, s \models \phi \iff s \in [\![\phi]\!]^{\mathcal{M}}$$

This can be read as *model $\mathcal{M}$ satisfies ϕ at state s*, or *ϕ is true at state s in model $\mathcal{M}$*.

Example 2. The core of this report is theorems and conjectures stating that all formulas that can be evaluated using a certain Kripke structure can also be evaluated using a different Kripke structure. Formally, this is stated as follows. Let $\mathcal{M}_1 = \{S_1, _, _\}$ and $\mathcal{M}_2 = \{S_2, _, _\}$ be two Kripke structures, and $f : S_1 \longrightarrow S_2$.

$$\forall \phi \in \mathcal{L}_n, \forall s \in S_1, \quad \mathcal{M}_1, s \models \phi \iff \mathcal{M}_2, f(s) \models \phi$$

A complete example can be found later in Example 11.

2.2 Multi-Agent Games with Imperfect Information Against Nature

This section relies on the definition used by Gurov et al. in [1] and [3].

2.2.1 Multi-Agent Games

Games are modeled using directed graphs, where the vertices represent the states of the game, and the edges represent the actions of the players (or agents).

Definition 2.2.1 (Multi-Agent Game with Imperfect Information Against Nature)**.**
A Multi-Agent Game with Imperfect Information Against Nature (MAGIIAN) is a tuple
$G = (Agt, Loc, l_I, \Sigma, \Delta, \mathcal{O})$, where:

- $Agt = \{1, ...n\}$ is a fixed team of n players (or agents)
- Loc is a finite set of locations (or states);
- $l_I \in Loc$ is the initial location;

- For each $i \in Agt$, Σ_i is the action alphabet of agent i;
- $\Sigma = \Sigma_1 \times ... \times \Sigma_n$ is a finite set of joint actions;
- $\Delta \subseteq Loc \times \Sigma \times Loc$ is a finite set of labeled transitions. It does not have to be the entirety of the cartesian product, it usually is only a subset (see the discussion below).
- For each $i \in Agt$, $\mathcal{O}_i$ is a partition of Loc in which each element corresponds to a possible observation by agent i. The states within a same possible observation are indistinguishable by the agent i.
- $\mathcal{O} = \mathcal{O}_1 \times ... \times \mathcal{O}_n$ is a finite set of joint observations.

Intuitively, Agt is a set of naturals, each acting as an identifier for a specific agent. Loc is the set of states in which the game can be from which l_I is the state of the game before any action is done. Each Σ_i is the set of actions each respective agent can perform, and Σ is the set of all combinations of individual actions. The structure of Σ allows any combination of actions to be done at any location. The joint actions available at a given location are instead captured by Δ. Given a location $l \in Loc$ and an action $\sigma \in \Sigma$, if σ can be performed at location l, then there exists $l' \in Loc$ such that $(l, \sigma, l') \in \Delta$. If the action cannot be performed at the location l, then there is no such l'. MAGIIAN are assumed to be played against an unpredictable environment (Nature). This is modeled by allowing for multiple such l' to exist. Finally, the games are with *imperfect information*. This means there is an equivalence relation between states that cannot be distinguished, which splits each $\mathcal{O}_i$ into blocks of various cardinals. Each of these blocks contains a possible observation. Given a location l and an agent i, $Obs_i(l)$ denotes the unique element of $\mathcal{O}_i$ such that $l \in Obs_i(l)$. Intuitively, $Obs_i(l)$ is the set of locations that agent i cannot distinguish from l.

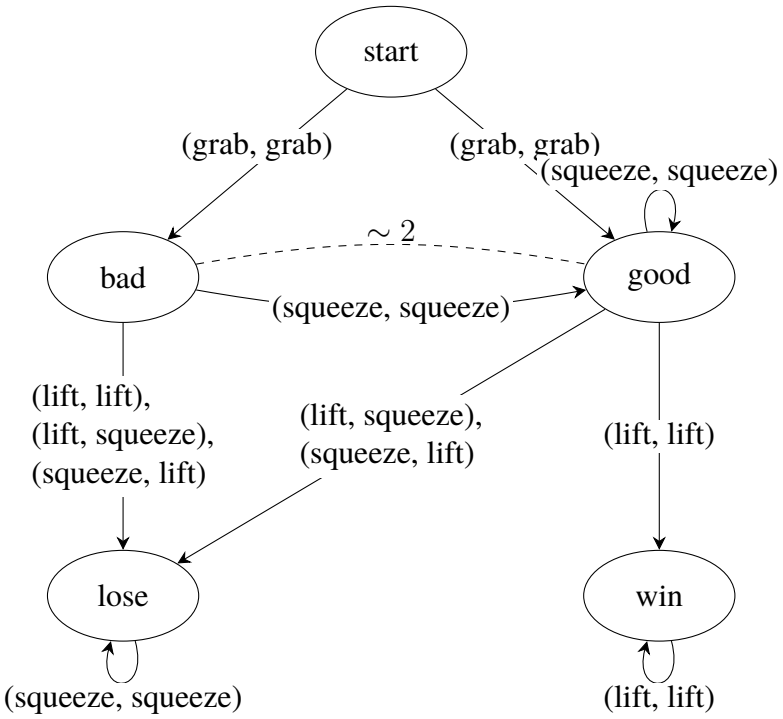


Figure 2.1: Cup-lifting game from [1]

Example 3. Example of the cup-lifting game from [1] represented in Figure 2.1. The game is played by two agents. Their objective is to successfully lift a cup. However, when they grab the cup, the grip might be either good or bad depending on the environment (Nature). One can imagine the cup being too slippery due to high humidity for instance. Agent 1 has a working grip sensor and can detect whether the grip is good or bad, whilst agent 2 has a defective grip sensor and cannot distinguish between a good and a bad grip. In order to win the game, they must both lift at the same time and when the grip is good.

- The game features a team of 2 agents $Agt = \{1, 2\}$
- The set of states is $S = \{start, bad, good, lose, win\}$.
- The initial state is $start$
- The sets of individual actions are identical for both agents: $\Sigma_1 = \Sigma_2 = \{grab, squeeze, lift\}$
- Transitions are defined as shown in Figure 2.1

- The observation partition for agent 1 is a partition consisting of a set of singletons. However, the observation partition for agent 2 contains the element $\{bad, good\}$ and singletons for other states, since agent 2 cannot distinguish between a good and a bad grip. This is represented by a dotted edge on Figure 2.1

This example will be used as a running example throughout the entire report.

Remarks.

- This definition can also be used to represent a game with *perfect information*. Such a game is characterized by the set of joint observations. If for each agent i, $\mathcal{O}_i$ is a set of singletons, then it is a game with *perfect information*. Similarly, the definition can also be used to represent a game that is not against Nature. In such a game, each state of the game can only have at most one outgoing transition for each joint-action.

- The sets considered in Def. 2.2.1 are assumed to be finite.

- Additionally, we assume it is possible to retrieve the current state of the game with absolute certainty by intersecting the observations of the agents. That is to say for any $l \in Loc$, $\bigcap_{i \in Agt} Obs_i(l) = \{l\}$. This means that there are not two distinct locations that are indistinguishable by all agents. This can be described as the *Perfect Distributed Knowledge* (or PDK for short) condition. Therefore, in this thesis any base game defined as a MAGIIAN is assumed to respect the PDK condition. MAGIIAN games that do not respect the PDK condition are out of scope for this thesis, but each use of the PDK condition will be made explicitly.

- Apart from the previous restrictions, we consider a general setting in which agents can only observe the state of the game and act, but cannot communicate with each other.

- The exact capability of agents playing a given game does not only depend on the definition of the game itself. The definition of the game dictates what actions are allowed, what are there consequences, and how agents can observe the state of the game. However, it does not capture the internal behaviour of an agent. That is to say, it does not

provide information on what each agent has in memory, or how they decide which action to take when multiple actions are allowed. This will instead be captured by the different models used to reason about the game. In this thesis, multiple models will be explored.

As the agent team Agt is only used as a means of indexing elements of Σ and $\mathcal{O}$, its definition is implicitly given by the definition of Σ and $\mathcal{O}$. Thus, we will omit it from game definitions and assume a team of n agents ($Agt = \{1, 2, ..., n\}$) for the rest of this thesis. This will simplify the notations without loss of generality.

Definition 2.2.2 (Indistinguishability Relations for Locations)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. For any agent i, define:

$$\forall l, l' \in Loc, l \sim_i l' \iff l \in Obs_i(l')$$

$\sim_i$ is the indistinguishability relation for agent i.

Note that by definition of $\mathcal{O}_i$ and Obs_i, the indistinguishability relations are equivalence relations.

Example 4. Using the running example of the cup-lifting game, we have:

$$bad \sim_2 good$$

Definition 2.2.3 (Plays and histories)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. A play is an infinite sequence $\pi = l_0\sigma_1 l_1 \sigma_2 l_2 ...$ such that $l_0 = l_I$ and for all $j \geq 0$, there exists $\sigma_j \in \Sigma$ such that $(l_j, \sigma_j, l_{j+1}) \in \Delta$.
A history is a finite prefix of a play and H_G denotes the set of all histories of the game. We additionally define the function $last$ by:

$$\begin{aligned} last : H_G &\longrightarrow Loc \\ l_I\sigma_1 l_1 ...\sigma_j l_j &\longmapsto l_j \end{aligned}$$

We finally define the length of an history as the number of locations visited in the history. More formally:

$$\mid l_I\sigma_1 l_1 ...\sigma_n l_n \mid = n + 1$$

Remarks.

- This thesis is restricted to synchronous games, i.e. all agents act simultaneously. This is captured by the previous definition of a play as an alternating sequence of locations and joint actions.

- The definition of a play allows for games to run infinitely. This is to define a mathematical object usable for any game (without termination constraint). Practically speaking, we will only use histories as we are interested in the state of the game at a given step and what happened beforehand, which is fully captured by the history.

Example 5.

$$h = start\ (grab, grab)\ good$$

is a valid history for the cup-lifting game. The game begins in the $start$ state, then agents perfom the joint action $(grab, grab)$ and the resulting state is $good$. Note that the history $h' = start\ (grab, grab)\ bad$ is also a valid history. The outcome of the joint action $(grab, grab)$ depends on *Nature*, hence the non-determinism of the outcome from the perspective of the agents.

We lift Def. 2.2.2 of indistinguishability relation to the histories. By abuse of notation, we will use the same symbol.

Definition 2.2.4 (Indistinguishability Relations for Histories)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. For any agent i, define:

$$\begin{aligned} & l_0\sigma_1 l_1 ... \sigma_k l_k \sim_i l'_0\sigma'_1 l'_1 ... \sigma'_{k'} l'_{k'} \\ \iff & k = k' \ \wedge \ \forall j \in \{0...k\}, (Obs_i(l_j) = Obs_i(l'_j) \wedge \sigma_j[i] = \sigma'_j[i]) \end{aligned}$$

Intuitively, an agent cannot distinguish two histories if it has taken the same actions and made the same observations (in the same order). For $k = k' = 0$, the previous definition gives the same relation as the indistinguishability relations for locations defined in Def. 2.2.2, which justifies the notation.

Example 6. Consider the following histories from the cup-lifting game from

Figure 2.1:

$h_1 =$ start (grab,grab) bad (squeeze, squeeze) good
$h_2 =$ start (grab,grab) good (squeeze, squeeze) good

Then $h_1 \sim_2 h_2$ but $h_1 \not\sim_1 h_2$.

2.2.2 Knowledge representation

As seen in Def. 2.1.1, epistemic logic involves reasoning on knowledge. Hence, we need to define proper terms and mathematical objects to reason about knowledge. Let us first look at an intuitive definition of higher-order knowledge. The *first-order* knowledge of an agent i corresponds to what agent i directly knows about the current state of the game. This can be represented by the set of states that agent i considers possible at a given step, when it only relies on its observation. For instance, at the beginning of the cup-lifting game from Figure 2.1, agent 2 knows that the game just started, so its first-order knowledge is $\{start\}$. After grabbing the cup, agent 2 knows that the game is either in state bad or $good$, this is its first-order knowledge after grabbing the cup.

The second-order knowledge is the knowledge each agent has about the first-order knowledge of each agent. For instance, assume the grip is good after grabbing the cup in the cup-lifting game. Then, agent 1 knows that agent 2 does not know whether the grip is $good$ or bad. Additionally, agent 1 knows that agent 1 knows that the grip is $good$, since its sensor is working properly. Note that this last sentence is the same as saying agent 1 knows that the grip is $good$, without the repetition.

Then, we can iterate this process. The *j-th*-order knowledge of an agent is the knowledge it has about the *(j-1)-th*-order knowledge of each agent. This relates to the notion of alternation depth defined after Def. 2.1.1. If a formula of alternation depth j is simply a chain of knowledge operators followed by an atomic proposition, then it is a statement about the *j-th*-order knowledge of the agent corresponding to the leftmost operator. For instance $K_i K_j K_l\ p$ can be viewed as a statement about the third-order knowledge of agent i.

To represent this intuitive notion of higher-order knowledge, we will use the knowledge structures as they result from iterated applications of the Multi-agent Knowledge-Based Subset Construction (MKBSC) defined in [1]. The details of the construction are not required to understand this report. The following definitions are presented by Gurov et al. in the unpublished

manuscript [3].

Definition 2.2.5 (Local and Global Knowledge Structures)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. For any knowledge order $j \geq 0$, the domains of *local knowledge structures* $A^{(j+1)}$ and *global knowledge structures* $B^{(j)}$ are defined by mutual recursion:

$$A^{(j+1)} = 2^{B^{(j)}}$$
$$B^{(0)} = Loc$$
$$B^{(j)} = (A^{(j)})^n$$

Intuitively, each element of $B^{(j)}$ is a vector with n coordinates, where n is the number of agents. The $i\text{-}th$ coordinate is an element of $A^{(j)}$ which can be used to represent the $j\text{-}th$-order local knowledge of agent i.

Example 7. Here are two examples of global knowledge structures from the cup-lifting game from Figure 2.1:

$$b_1 = (\{good\}, \{bad, good\})$$
$$b_2 = \Big(\big\{(\{good\}, \{bad, good\})\big\},$$
$$\big\{(\{good\}\ ,\ \{bad, good\}), (\{bad\}, \{bad, good\})\big\}\Big)$$

$b_1 \in B(1)$ is a first order global knowledge structure whilst $b_2 \in B^{(2)}$ is a second order global knowledge structure. Each element of vector b_1 is a first-order local knowledge structure, whilst each element of vector b_2 is a second-order local knowledge structure.

Remark. As knowledge structures result from iterated applications of the MKBSC, there is the following consistency constraint: For all $j > 0$ and $b \in B^{(j+1)}$:

$$\forall i \in Agt,\ \forall b_1, b_2 \in b[i],\ b_1[i] = b_2[i]$$

Now that a knowledge representation is defined, wee need to define how to use it. In multi-agent games, it is important for the agent to be able to update the knowledge during the play. Otherwise, their behaviours would remain the same independently of the state of the game. Hence the need for an update function.

2.2.3 Knowledge Update

Definition 2.2.6 (Knowledge Update Functions)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. The *local knowledge update functions* of agent i, $\delta_i^j : A^{(j)} \times \Sigma_i \times \mathcal{O}_i \longrightarrow A^{(j)}$ is defined by induction on the knowledge order:

$$\delta_i^1(a, \sigma_i, o_i) = \Big\{ l' \in o_i \mid \exists \sigma \in \Sigma. \big(\sigma[i] = \sigma_i \wedge \exists l \in a. (l, \sigma, l') \in \Delta\big) \Big\}$$

$$\delta_i^{j+1}(a, \sigma_i, o_i) = \Big\{ \big(\delta_{i'}^j\,(b[i'], \sigma[i'], o[i'])\big)_{i' \in Agt} \mid b \in a, \sigma \in \Sigma, \sigma[i] = \sigma_i, o \in \mathcal{O}, o[i] = o_i \Big\}$$

The *global knowledge update function* $\delta^j : B^{(j)} \times \Sigma \times \mathcal{O} \longrightarrow B^{(j)}$ is defined as the vector of the local updates:

$$\delta^j(b, \sigma, o) = \big(\delta_i^j(b[i], \sigma[i], o[i])\big)_{i \in Agt}$$

These functions are used to maintain an up to date version of the local knowledge structure. The process of agent i using the first-order local knowledge update function can be described as follow:

1. Agent i as a local knowledge a. It contains all the state that agent i considers possible.
2. Agent i makes an individual action σ_i. Each agent made a simultaneous individual action, but agent i does not know what these actions are.
3. Agent i makes an observation o_i about the game.
4. Agent i retrieve all the joint actions that contain σ_i.
5. For each joint action retrieved in the previous step, agent i checks if the action is allowed. That is to say, for each state that agent i considered possible in step 1, it checks if the joint action can be taken from the state. If so the state is added to a temporary set.
6. The updated first-order local knowledge structure is exactly the intersection between the temporary set and the observation o_i made in step 3.

Example 8. Let us take an example from the cup-lifting game shown in Figure 2.1. Let us consider agent 2, starting from the initial state of the game.

1. The initial first-order local knowledge structure of agent 2 is $\{start\}$.
2. Agent 2 makes the individual action $\sigma_2 = grab$. At this step, agent 2 does not know what agent 1 did.
3. Agent 2 observes the state of the game $o_2 = \{good, bad\}$.
4. Agent 2 retrieves the joint actions $(grab, grab)$, $(lift, grab)$ and $(squeeze, grab)$. These are the only three actions that contain $grab$ as a second member.
5. The action $(grab, grab)$ can be taken from the state $start$ which is the only possible state in step 1. The resulting state is either $good$ or bad. Both are added to a temporary set $S_t = \{good, bad\}$. The actions $(lift, grab)$ and $(squeeze, grab)$ are not actions that can be taken from the $start$ location (cf Figure 2.1). Hence, no additional element is added to the temporary set.
6. The updated first-order local knowledge structure of agent 2 is $S_t \cap o_2 = \{good, bad\}$.

If we play one more step by saying that both agents squeeze the cup, then the state of the game is $good$. We then have $o_2 = \{good, bad\}$, since agent 2 cannot distinguish between these two states. Then the two possible joint actions are $(squeeze, squeeze)$ and $(lift, squeeze)$, and $S_t = \{good, lose\}$ (respective results of these two actions). Finally, the intersection is $good$, so agent 2 now know that the grip is good.

This example was for first-order local knowledge structures. For higher-order local knowledge structures, the process is done recursively since each element of a *(j+1)-th*-order local knowledge structure is a *j-th*-order local knowledge structure.

We then define the lifted knowledge update functions on histories. The lifted update functions are used to simplify the notations and not write recursive calls for each element of the history.

Definition 2.2.7 (Lifted Knowledge Update Functions)**.**

$$\hat{\delta}_i^j(l_0) = Init_i^j(l_0)$$

$$\hat{\delta}_i^j(l_0\sigma_1 l_1 ... \sigma_m l_m \sigma_{m+1} l_{m+1}) = \delta_i^j\left(\hat{\delta}_i^j(l_0\sigma_1 l_1 ... \sigma_m l_m), \sigma_{m+1}[i], Obs_i(l_{m+1})\right)$$

Where Init functions are defined by induction on the knowledge order as follows:

$$Init_i^1(l) = Obs_i(l)$$
$$Init_i^{j+1} = \{(Init_{i'}^j(l))_{i' \in Agt}\}$$

The *global knowledge update function* on histories $\hat{\delta}^j : H_G \longrightarrow B^{(j)}$ is defined as the vector of the local updates on histories:

$$\hat{\delta}^j(h) = (\hat{\delta}_i^j(h))_{i \in Agt}$$

Example 9. Example from the cup-lifting game from Figure 2.1. Let $h = start\ (grab, grab)\ bad$:

$$\hat{\delta}_1^1(h) = \{bad\}$$
$$\hat{\delta}_2^1(h) = \{bad, good\}$$
$$\hat{\delta}^1(h) = (\{bad\}, \{bad, good\})$$

Intuitively, each element of $\hat{\delta}^1(h)$ corresponds to the first-order *local* knowledge of the respective agent. Hence, it contains the current possible states from the point of view of each agent. It is a subset of the agent's observation, as highlighted by Example 8.

For the second-order knowledge structures, we have:

$$\hat{\delta}_1^2(h) = \Big\{ \big(\{bad\}, \{bad, good\} \big) \Big\}$$
$$\hat{\delta}_1^2(h) = \Big\{ \big(\{bad\}, \{bad, good\} \big), \big(\{good\}, \{bad, good\} \big) \Big\}$$
$$\hat{\delta}^2(h) = \Big(\big\{ (\{good\}, \{bad, good\}) \big\}, \big\{ (\{good\}\ ,\ \{bad, good\}), (\{bad\}, \{bad, good\}) \big\} \Big)$$

Intuitively, each element of $\hat{\delta}^2(h)$ corresponds to the second-order *local* knowledge of the respective agents. By definition of the knowledge structure, each element is the set of first-order *global* knowledge structures that each agent considers possible. For instance, since agent 2 cannot distinguish whether the state is $good$ or bad, then the second element contains two alternatives: either agent 1 knows that the state is $good$, or agent 1 knows that the state is bad. In both cases, agent 2 cannot decide between $good$ and bad.

Using the MKBSC-expansion defined by Gurov et al. in [1], we define expanded games. An expanded game is played on knowledge structures instead of game locations.

Definition 2.2.8 (Expanded Game)**.**
Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN. Given a knowledge order $j \in \mathbb{N}$ the j-expansion is defined by:

$$G^{0K} = G$$
$$G^{jK} = (S^{jK}, s_I, \Sigma, \Delta^{jK}, \mathcal{O}^{jK})$$

where:

$$S^{jK} = \{\hat{\delta}^j(h) \mid h \in H_G\}$$
$$s_I = Init^j(l_I)$$
$$s \in Obs_i^{jK}(s') \iff s[i] = s'[i]$$
$$(s, \sigma, s') \in \Delta^{jK} \iff \delta^j(s, \sigma, Obs^{jK}(s')) = s'$$

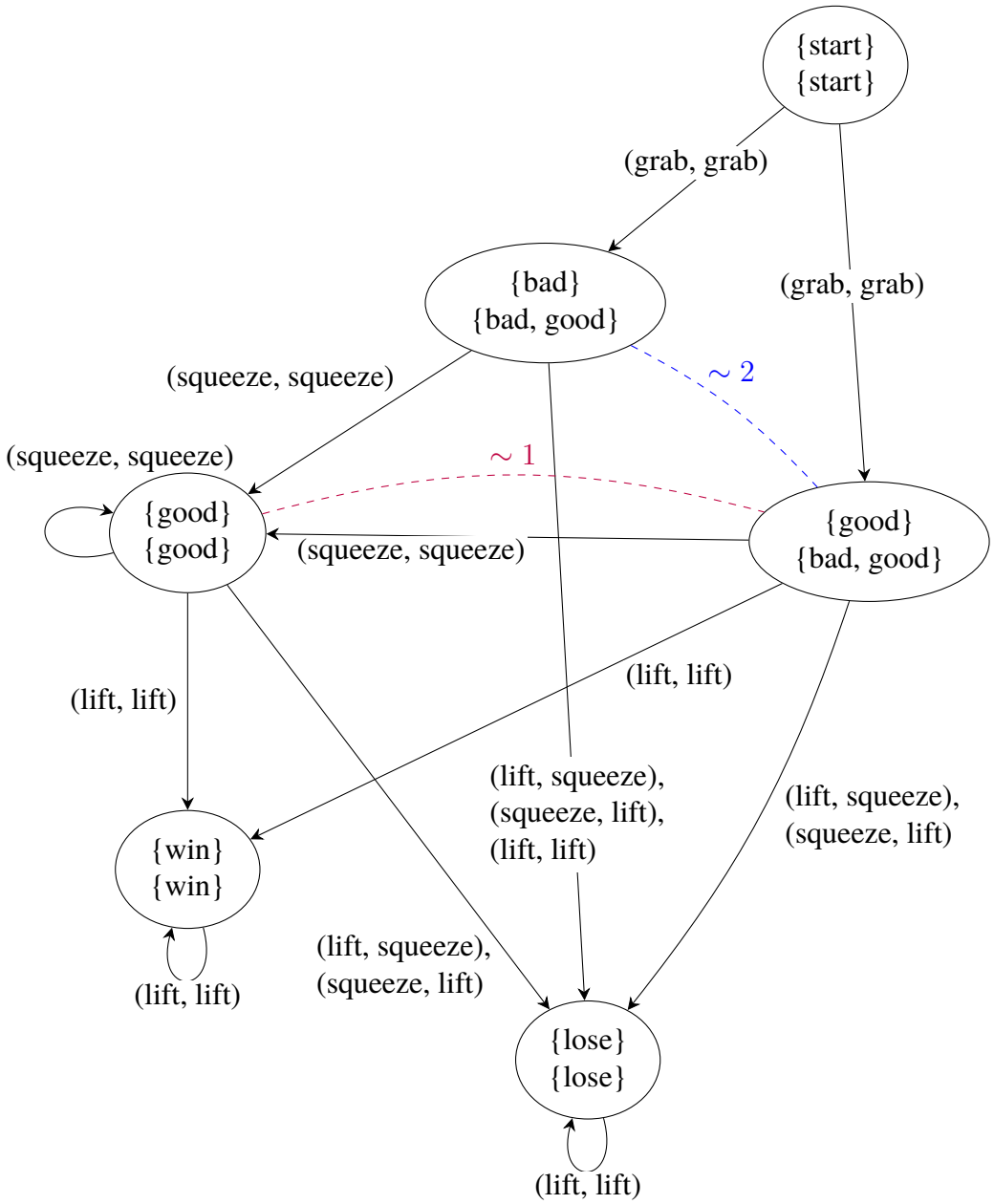


Figure 2.2: First expansion of the cup-lifting game from [1]

Example 10. Let G be the cup-lifting game as described in Figure 2.1. The first expansion G^{1K} of G is as pictured in Figure 2.2. For expanded games, the possible observations of agent i are groups of states for which the local knowledge structure of i is the same. For the cup-lifting base game, only the states bad and $good$ are part of the same observation block for agent 2. However, for the first expansion of the game, there are two states

indistinguishable by agent 1 and two states indistinguishable by agent 2. These indistinguishable states are linked by dashed edges in Figure 2.2: red for agent 1, and blue for agent 2.

Remark. The condition $\forall l \in Loc, \bigcap_{i \in Agt} Obs_i(l) = \{l\}$ we assumed previously for base games ensures the *Perfect Distributed Knowledge* condition (or PDK for short) is met for any expanded game [1]. Every use the PDK condition in this report will be made explicitly.

Now that the logic has been defined in 2.1, and basic notions related to MAGIIAN have been introduced, we can look at how to evaluate formula.

2.2.4 History-Based Interpretation

There are multiple ways of interpreting epistemic formulas for a MAGIIAN. To do so, one need to define a Kripke structure that derives from the game. The most intuitive and standard approach is to rely on histories.

Definition 2.2.9 (History-Based Kripke Structure)**.** Let $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ be a MAGIIAN, and H_G the set of all possible histories of G. This induces a Kripke structure $\mathcal{M}_{H_G} = (H_G, \{\sim_i\}_{i \in Agt}, L)$ where:

- The $\{\sim_i\}_{i \in Agt}$ are the indistinguishability relations
- L is the labelling function defined by:

$$\begin{aligned} L : H_G &\longrightarrow 2^{\mathcal{AP}} \\ h &\longmapsto \{p_{last(h)}\} \end{aligned}$$

where $p_{last(h)}$ denotes the atomic proposition true at location $last(h)$

Intuitively, this model captures and retain all pieces of information about the game. This ensures any formula can be evaluated but come with a high cost in terms of complexity. As the game goes on, larger and larger histories need to be accounted for which become expansive both space-wise and time-wise. Hence, the need for a sparser model, in the sense that it does not contain the whole histories, but still captures and retain sufficient information for this evaluation of epistemic formulas. When introducing new interpretations in chapter 4, we will always compare them to this standard approach. The objective is for other interpretations introduced in chapter 4 to be usable to evaluate any formula that can be evaluated with the standard interpretation.

Example 11. Let G be the cup-lifting game from Figure 2.1. Let $h = start\ (grab, grab)\ bad$, and $\phi = K_1 K_2\ ((K_1 good) \vee (K_1 bad))$.
In natural language, $\psi = K_2\ ((K_1 good) \vee (K_1 bad))$ this means that agent 2 knows that agent 1 knows the current state of the game, be it good or bad. Thus, $\phi = K_1 \psi$ means that agent 1 knows what is previously mentioned.
Let us evaluate this formula using the history-based interpretation. First, we consider the leftmost knowledge operator:

$$\begin{aligned} &\mathcal{M}_{H_G}, h \models K_1 \psi \\ &\iff \forall h' \in H_G, h' \sim_i h \implies \mathcal{M}_{H_G}, h' \models \psi \end{aligned}$$

Therefore, we need to identify all histories that agent 1 cannot distinguish from h. Looking at Figure 2.1, the only such element is h itself. We can then look at the rest of the formula:

$$\begin{aligned} &\mathcal{M}_{H_G}, h \models K_2\ (K_1 good \vee K_1 bad) \\ &\iff \forall h' \in H_G, h' \sim_2 h \implies \mathcal{M}_{H_G}, h' \models (K_1 good) \vee (K_1 bad) \end{aligned}$$

This time, we need to identify all histories that agent 2 cannot distinguish from h. Since agent 2 cannot distinguish between $good$ and bad, we find one additional matching history: $h_0 = start\ (grab, grab)\ good$. Hence, we now need to evaluate the subformula $(K_1 good) \vee (K_1 bad)$ for h and h_0. Since agent 1 can distinguish between $good$ and bad, $K_1 good$ evaluates to true at h_0, and $K_1 bad$ evaluates to true at h. Thus, $K_1 good \vee K_2 good$ evaluates to true at both h and h_0. Therefore, the complete formula evaluates to true at h, and we have:

$$\mathcal{M}_{H_G}, (start\ (grab, grab)\ bad) \models K_1 K_2\ ((K_1 good) \vee (K_1 bad))$$

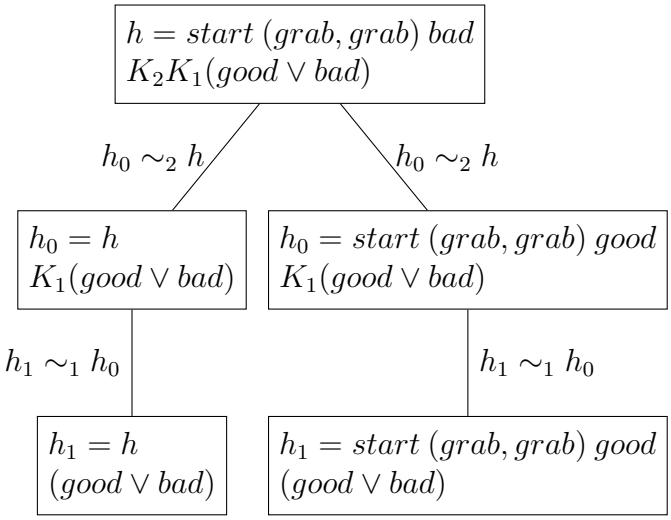
2.3 Visual Representation of Formula Evaluation

In order to simplify examples of evaluations of formulas such as the previous one from Example 11, we introduce a tree-like visualisation for formulas with sequence of knowledge operators.

Each node of the tree is a pair of a point where the evaluation is made, and the formula that must be evaluated. If the leftmost element of the formula is a

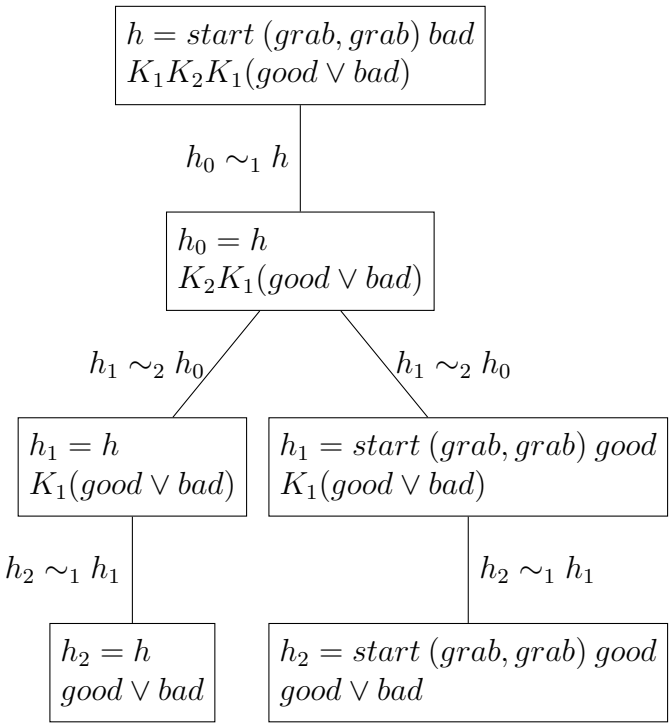
knowledge operator, then branches go from the node to children nodes. Each branch is labeled with the accessibility relation, and each child corresponds to the accessible points, and the sub-formula to be verified (which is the same as the parent, without the leftmost knowledge operator). The same process is applied until the leaves contain formulas without knowledge operators. Once the tree is constructed, the result of the evaluation is the conjunction of the evaluations of the formula of each leaf at their respective point.

Example 12.
The evaluation tree for the evaluation from Example 11 is:



To finish the evaluation, one need to verify $\mathcal{M}_{H_G}, h \models good \vee bad$ and $\mathcal{M}_{H_G}, start\ (grab, grab)\ good \models good \vee bad$. Both statements hold, since $last(h) = bad \in \{good, bad\}$ and $good \in \{good, bad\}$.

Example 13.
The evaluation tree for the evaluation from Example 11 is:



To finish the evaluation, one need to verify $\mathcal{M}_{H_G}, h \models good \vee bad$ and $\mathcal{M}_{H_G}, start\ (grab, grab)\ good \models good \vee bad$. Both statements hold, since $last(h) = bad \in \{good, bad\}$ and $good \in \{good, bad\}$.

2.4 Knowledge-Based Programs

In this section, we will provide a brief introduction to knowledge-based programs (KBP), as they are described by Fagin et al. in [4]. This notion is not required to understand the core part of this thesis. However, it provides a justification for this theoretical work, as well as a good illustration of practical considerations discussed at the end of this report. A knowledge-based programs for agent i is a program with the following form:

begin KBP_i
  if $t_1 \wedge K_i\phi_1$ **do** a_1
  if $t_2 \wedge K_i\phi_2$ **do** a_2
  ...
end

Where each t_j is a standard test, each ϕ_j is a formula of epistemic logic, and each a_j is an action made by agent i. By standard test, one must understand a test as it could be done in imperative programming.

Example 14. Let us take an example from the cup-lifting game. To also illustrate what a standard test can be, let us consider that robot 2 checks its $battery_level$ percentage to decide whether it can grab the cup or not. The first line of the knowledge-based program could look like this:

begin KBP_i
  if $(battery_level \geq 20) \wedge K_2\, start$ **do** $grab$
  ...
end

In multi-agent games, we generally define one knowledge-based program per agent. To actually run such programs, one need to be able to evaluate epistemic logic formulas.

From the syntax of the epistemic logic and its sub-logic with bounded alternation depth (Def. 2.1.1), we can define two different groups of knowledge-based programs. The first group consists of knowledge-based programs that involve unbounded alternation depth. The second group consists of programs that involve only formulas of epistemic logic with a bounded alternation depth.

The practical consequences of the results of this thesis will be briefly discussed by using these two groups of knowledge-based programs.

2.5 Related Work

In [1] Gurov et al. introduce MKBSC and knowledge structures that derive from its iterated application. In the unpublished manuscript [3], Gurov et al. define different Kripke structures that rely on knowledge structures, and claim they can be used to evaluate formula of $\mathcal{L}_n$ or one of its sub logic. These are the conjectures that will be proven in this thesis. In order to simplify the reading of this report and avoid numerous reference to this part of the thesis, these definitions and conjectures will not be stated in this section. They will be directly integrated in the result section, with proper credit.

Additionally, bachelor and master thesis supervised by Gurov investigates different questions related to MAGIIAN. In [5], Gustaf Pihl investigate knowledge representation for MAGIIAN especially in the context of stable

games. In [6] Carl Åkerman Rydbeck formalizes knowledge for MAGIIAN using epistemic logic.

Another knowledge representation has been introduced by van der Meyden in [2]. In [2], van der Meyden introduces k-trees to represent knowledge for a sub-logic of epistemic-logic where the *alternation depth* is bounded by k. They serve the same purpose as *knowledge structures*, which is to abstract from histories (or traces in case of [2]) and use them store and update knowledge of agents throughout the game, with the evaluation of epistemic formulas as the final goal. This requires to retain enough information, but also to be able to update the representation without recomputing it entirely at each step. Note that k-trees originate from an effort of abstraction from the history, whilst knowledge structures originate from the MKBSC algorithm. In [2], van der Meyden proves the iteratively computed k-trees can be used to evaluate any formula $\phi \in \mathcal{L}_n$ such that ϕ as an alternation depth of at most k. The detailed definitions and theorems will not be integrated to this report as they are not necessary for understanding. It is however important to note certain points. First, van der Meyden's k-trees are limited to a sub logic of epistemic logic. Whilst our goal is to explore different representations in order to pave the way for a model that can handle formulas with unbounded alternation depth. Second, it is important noting that given a history, a knowledge structure contains exactly the leaves of the corresponding k-tree. From this second observation, we can give a second definition for knowledge structures inspired by van der Meyden's work.

Definition 2.5.1 (Unwinding for Knowledge Structures)**.** The *global knowledge structure unwinding* F''_j and local knowledge structure unwinding $U''_{j,i}$ are defined by mutual recursion:

$$\begin{aligned} F''_j &: H_G \longrightarrow B^{(j)} \\ U''_{j,i} &: H_G \longrightarrow A^{(j)} \\ F''_0(h) &= last(h) \\ U''_{j,i}(h) &= \{F''_j(h') \mid h' \sim_i h\} \\ F''_{j+1}(h) &= (U''_{j,i}(h))_{i \in Agt} \end{aligned}$$

Lemma 1. *Let* $G = (Loc, l_I, \Sigma, \Delta, \mathcal{O})$ *be a MAGIIAN.*

$$\forall h \in H_G, \hat{\delta}^j(h) = F''_j(h)$$

The proof follows directly from Lemma 4.6 from van der Meyden's paper [2] and the second observation from above. Note that this definition makes

use of the indistinguishability relation, which is an equivalence relation. In [2], the accessibility relation used for k-trees is not symmetric.

Example 15. Let us take an example from the cup-lifting game. To illustrate both the computation of the unwindings and Lemma 1, let us consider the same history we used to illustrate knowledge update functions lifted to histories in Example 9. Let $h = start\ (grab, grab)\ bad$. First, we compute the global unwinding $F''_0(h)$:

$$F''_0(h) = last(h) = bad$$

Then, we can compute the first-order local unwindings. For agent 1, we have:

$$\begin{aligned} U''_{0,1}(h) &= \{F''_0(h') \mid h' \sim_1 h\} \\ &= \{F''_0(h)\} \\ &= \{bad\} \\ &= \hat{\delta}^1_1(h) \qquad \textit{(cf Example 9)} \end{aligned}$$

Since agent 1 can distinguish bad from any other state. For agent 2, we have:

$$\begin{aligned} U''_{0,2}(h) &= \{F''_0(h), F''_0(\ start(grab, grab)good\ )\} \\ &= \{bad, good\} \\ &= \hat{\delta}^1_2(h) \qquad \textit{(cf Example 9)} \end{aligned}$$

Putting these two results in order gives us the first-order global unwinding:

$$F''_1(h) = (\{bad\}, \{bad, good\}) = \hat{\delta}^1(h) \qquad \textit{(cf Example 9)}$$

To compute the second-order global unwinding, we will first need to compute $F''_1\big(start(grab, grab)good\big)$. This can easily be done by following the previous computation. By analogy, the result is the same as $F''_1(h)$, but with

the first element replaced by $\{good\}$. Then, we can compute $F_2''(h)$:

$$
\begin{aligned}
F_2''(h) &= \Big(\{F_1''(h') \mid h' \sim_1 h\}, \{F_1''(h') \mid h' \sim_2 h\} \Big) \\
&= \Big(\{F_1''(h)\}, \big\{F_1''(h), F_1''\big(start(grab, grab)good\big)\big\} \Big) \\
&= \Big(\{(\{bad\}, \{bad, good\})\}, \\
&\qquad \big\{(\{bad\}, \{bad, good\}), F_1''\big(start(grab, grab)good\big)\big\} \Big) \\
&= \Big(\{(\{bad\}, \{bad, good\})\}, \\
&\qquad \{(\{bad\}, \{bad, good\}), (\{good\}, \{bad, good\}))\} \Big) \\
&= \hat{\delta}^2(h) \qquad \textit{(cf Example 9)}
\end{aligned}
$$

Chapter 3

Method and Mathematical Tools

The purpose of this chapter is to provide an overview to the method and mathematical tools used in this thesis. The results of this thesis are mathematical proofs of theorems conjectured by Gurov et al. in [3]. Thus, this report relies on formal mathematical concepts and methods. To understand the required reasoning, section 3.1 describes reasoning on epistemic logic using structural induction, and section 3.2 introduces the main mathematical tools used to reason on knowledge structures.

3.1 Reasoning on Epistemic Logic

The syntax of epistemic logic is recursively defined in Def. 2.1.1, and so is the semantics in Def. 2.1.3. Thus, most proofs aiming to show a theorem holds for any formula will be done by structural induction. That is to say proving the base cases, which are for $\phi = X$ and $\phi = p$, then proving the inductive steps hold assuming an induction hypothesis. The inductive steps are the other derivations allowed by Def. 2.1.1: $\phi = \neg\phi_1$, $\phi = \phi_1 \vee \phi_2$, and $\phi = K_i\phi_1$. The first two steps will be direct rewriting using the semantics. The third step requires more careful considerations.

3.2 Reasoning on Knowledge Structures

The formal definition of knowledge update functions (Def. 2.2.6) is heavy and introduces numerous symbols. Thus, when reasoning on knowledge structures is required, we will instead rely on two other methods (when possible). The first approach is to use the unwindings since by Lemma 1: $\hat{\delta}^j = F''_j$ for any

$j > 0$. The definition of the knowledge update functions is done by induction on the length of the history, and by induction on the knowledge order. On the other hand, the definition of the unwindings is done directly by induction on the knowledge order. Thus, using the knowledge update functions would require an additional proof by induction on the length of the history, compared to a proof made using the unwindings. The second approach is by analogy with existing representations, such as van der Meyden's k-trees defined in [2]. This has been used previously for the proof of Lemma 1. Such analogies in the proofs will only be kept to minimal so this report can be read on its own. Regardless of which approach is chosen, a tool that can be used is proof by induction on the knowledge order, especially since the definitions of update functions and their unwindings are done by induction.

Chapter 4

Results

As mentioned in the background chapter, there are multiple ways of interpreting epistemic formulas for a MAGIIAN. History-based Kripke structures defined in Def. 2.2.9 is the standard approach, which relies on histories. In the following sections, other Kripke structures will be defined with the objective of being able to evaluate any formula that can be evaluated using the history-based approach. As briefly presented in Example 2, this will result in conjectures and theorems of the same general shape. The final objective is to obtain a representation that can not only be used to express any formula, but also be used in a practical setting with the restrictions presented in section 2.2, such as the impossibility for agents to communicate. In this thesis, we define and discuss three different interpretations.

4.1 Interpretations Based on Knowledge Structures

The first alternative Kripke structure is based on knowledge structures. The following Kripke structure is introduced by Gurov et al. in [3]. Recall the definitions of local and global knowledge structures domains $A^{(j+1)}$ and $B^{(j)}$ in Def. 2.2.5. For any knowledge structure $b \in B^{(j)}$, $\downarrow b$ denotes the downward closure, also called the lower closure, of b.

Definition 4.1.1 (Knowledge Structure-Based Kripke Structure)**.**
Assume the PDK holds. Let $b \in B^{(j)}$ be a global knowledge structure of order j. It induces a Kripke structure $\mathcal{M}_b = (\downarrow b, \{\rightarrow_i\}_{i \in Agt}, L)$ where:

- For all $b_1, b_2 \in \downarrow b$, we define $b_1 \rightarrow_i b_2 \iff b_2 \in b_1[i]$
- For all $b' \in \downarrow b$, we define $L(b') = \{p_{\Cap b'}\}$

where $\Cap b'$ is the only element of the iterated intersection of elements of the tuples in b'. By the PDK condition, $\Cap b'$ is a singleton. So by abuse of notation we take its single element as the result, and $p_{\Cap b'}$ is well-defined.

Remark. The labelling function is only well-defined if the PDK holds. Refer to remarks on Def. 2.2.8 for more details on the PDK condition.

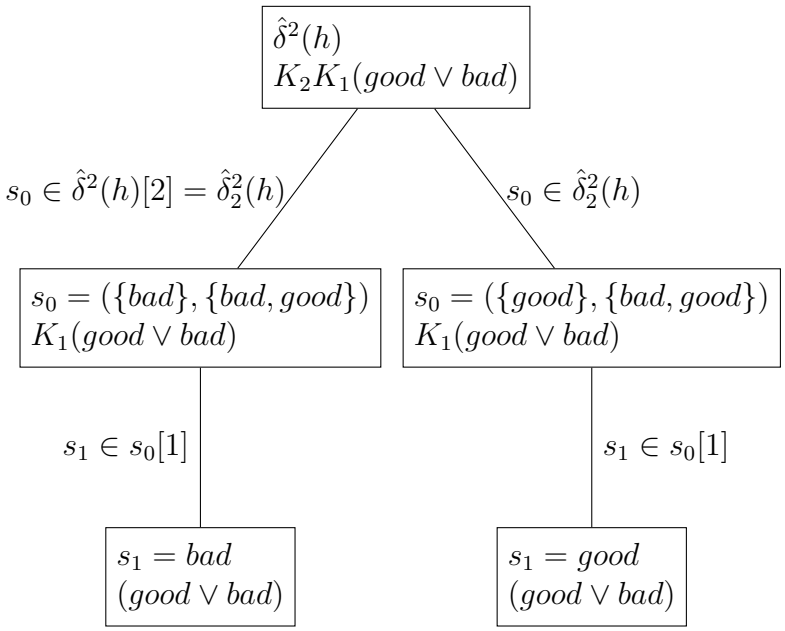
Example 16.
Let us take an example from the cup-lifting game. Let $h = start\ (grab, grab)\ bad$ and $\phi = K_2K_1(bad \vee good)$. Then the second order global knowledge is:

$$b = \hat{\delta}^2(h) = \Big(\big\{\ (\{bad\}, \{bad, good\})\ \big\} \\ \big\{\ (\{bad\}, \{bad, good\}),\ (\{good\}, \{bad, good\})\ \big\}\Big)$$

We can then try to evaluate $\mathcal{M}_{\hat{\delta}^2(h)}\hat{\delta}^2(h) \models K_2K_1(good \vee bad)$. The first evaluation step consists of evaluating the leftmost knowledge operator. By definition of the accessibility relation for the knowledge structure-based kripke structure, we have:

$$\mathcal{M}_{\hat{\delta}^2(h)}\hat{\delta}^2(h) \models K_2K_1(good \vee bad) \\ \iff \forall s_0 \in \downarrow b, s_0 \in \hat{\delta}^2(h) \implies \mathcal{M}_{\hat{\delta}^2(h)}, s_0 \models K_1(good \vee bad)$$

There are two points s_0 such that $s_0 \in \hat{\delta}^2(h)$. Either $s_0 = (\{bad\}, \{bad, good\})$, or $s_0 = (\{good\}, \{bad, good\})$. We then repeat the same process to evaluate $K_1(good \vee bad)$ for each of these points. This leads to the following evaluation tree:



Finally, for the left leaf we have $bad \in \{good, bad\}$ and for the right leaf we have $good \in \{good, bad\}$. Therefore each leaf evaluates to true and the whole formula evaluates to true.

The previous definition allows for interpretation using the semantics of *epistemic logic* previously defined in Def. 2.1.3 for formula evaluation using Kripke structures. One can also define another satisfaction relation to evaluate formulas directly over knowledge structures, without using Kripke structures. This alternative semantics is introduced in [6].

Definition 4.1.2 (Semantics of $\mathcal{L}_n^{jK}$ for Knowledge Structures)**.**
Assume the PDK holds. We define the semantics by inductively defining the satisfaction relation for any local knowledge structure a and global knowledge structure b as follows:

$$
\begin{aligned}
b \models_B p_l &\iff \Cap b = l \\
b \models_B \neg\phi &\iff not\,(b \models_B \phi) \\
b \models \phi_1 \vee \phi_2 &\iff (b \models_B \phi_1) \vee (b \models_B \phi_2) \\
b \models_B K_i\phi &\iff b[i] \models_A \phi
\end{aligned}
$$

$$
a \models_A \phi \iff \forall b \in a, b \models_B \phi
$$

Remarks.

- Once again, the PDK condition must be satisfied for the semantics to be well-defined.
- The semantics is not defined on the whole *epistemic logic* $\mathcal{L}_n$, but only on $\mathcal{L}_n^{jK}$ which is a subset of *epistemic logic* where formulas are of alternation depth at most j (see Def. 2.1.1).

This semantics relies on knowledge structures only, which is motivated by knowledge-based programs and strategies. It differs from the history-based interpretation that relied on Kripke structures, but its expressive power is the same. This is the object of the following theorem featured as the third conjecture in Gurov et al. manuscript [3].

Theorem 1. *For any* $\phi \in \mathcal{L}_n^{jK}$ *and* $b \in B^{(j)}$*:*

$$\mathcal{M}_b, b \models \phi \iff b \models_B \phi$$

Proof of Theorem 1.
The proof is done by structural induction on $\phi \in \mathcal{L}_n^{jK}$.

- Base case:

$$\begin{aligned}
&\mathcal{M}_b, b \models p_l \\
&\iff p_l \in L(b) && \textit{(Semantics)} \\
&\iff l = \Cap b && \textit{(PDK + labelling function)} \\
&\iff b \models_B p_l && \textit{(Semantics)}
\end{aligned}$$

- Induction step:

  - Step $\neg\phi$:
    Assume ϕ of alternation depth at most j and:

$$\mathcal{M}_b, b \models \phi \iff b \models_B \phi$$

    Then, by definition of alternation depth, $\neg\phi$ is also of alternation

depth at most j (see comments following Def. 2.1.1), and:

$$\begin{aligned}
&\mathcal{M}_b, b \models \neg\phi \\
&\iff b \in B^{(j)} \setminus [\![\phi]\!]^{\mathcal{M}_b} && \textit{(Satisfaction Relation)} \\
&\iff b \in B^{(j)} \wedge not(b \in [\![\phi]\!]^{\mathcal{M}_b}) && \textit{(Semantics Def. 2.1.3)} \\
&\iff not(\mathcal{M}_b, b \models \phi) && \textit{(Satisfaction Relation)} \\
&\iff not(b \models_B \phi) && \textit{(Induc. hyp.)} \\
&\iff b \models_B \neg\phi && \textit{(Semantics Def. 4.1.2)}
\end{aligned}$$

Therefore, the induction step for $\neg\phi$ holds.

– Step $\phi \vee \phi$:
Assume $\phi_1,\ \phi_2$ of alternation depth at most j and $\forall\psi \in \{\phi_1, \phi_2\}$:

$$\mathcal{M}_b, b \models \psi \iff b \models_B \psi$$

Then, by definition of alternation depth, $\phi_1 \vee \phi_2$ is also of alternation depth at most j (see comments following Def. 2.1.1), and:

$$\begin{aligned}
&\mathcal{M}_b, b \models \phi_1 \vee \phi_2 \\
&\iff (\mathcal{M}_b, b \models \phi_1) \vee (\mathcal{M}_b, b \models \phi_2) \\
&\iff (b \models_B \phi_1) \vee (b \models_B \phi_2) \\
&\iff b \models_B \phi_1 \vee \phi_2
\end{aligned}$$

Therefore, the induction step for $\phi \vee \phi$ holds.

– Step $K_i\phi$:
Assume ϕ of alternation depth at most $j - 1$ and:

$$\mathcal{M}_b, b \models \phi \iff b \models_B \phi$$

Then $K_i\phi$ is of alternation depth at most j and we have:

$$
\begin{aligned}
&\mathcal{M}_b, b \models K_i\phi \\
&\iff \forall b' \in \downarrow b, \\
&\qquad (b \to_i b' \implies \mathcal{M}_b, b' \models \phi) && \textit{(Def. 4.1.1)} \\
&\iff \forall b' \in \downarrow b, \\
&\qquad (b' \in b[i] \implies b' \models_B \phi) && \textit{(Induc. hyp.)} \\
&\iff \forall b' \in b[i], b' \models_B \phi && \textit{(By def. of} \downarrow b\textit{:, } b[i] \subseteq \downarrow b\textit{)} \\
&\iff b[i] \models_A \phi && \textit{(Semantics)} \\
&\iff b \models_B K_i\phi && \textit{(Semantics)}
\end{aligned}
$$

□

Note that the proof of the induction step requires recursive accesses to $b[i]$, which is the reason why the alternation depth must be bounded by the knowledge order. If the alternation depth were greater than the knowledge order, at the j+1 alternation we would get $b \in B^{(0)}$, so b would be a singleton and $b[i]$ would be undefined.

Now that these two representations have been proven to be equivalent, the equivalence with the standard history-based representation must be proven. This theorem is also featured as a conjecture in Gurov et al. manuscript [3]. Recall Def. 2.5.1 of F'', and Lemma 1. Instead of directly using knowledge structures $\hat{\delta}$, we will use their unwindings to simplify the proof.

Theorem 2. *Let G be a MAGIIAN and $\mathcal{M}_{H_G}$ be the history-based Kripke structure induced by H_G. Let $h \in H_G$ be a history of G, $\phi \in \mathcal{L}_n^{jK}$ be a formula with alternation depth at most j:*

$$\mathcal{M}_{H_G}, h \models \phi \iff F''_j(h) \models_B \phi$$

Before proving this theorem, let us first prove the following lemma, which helps understand the base case.

Lemma 2. *Under the PDK assumption, for any history h and any knowledge-order j:*

$$\Cap F''_j(h) = last(h)$$

Proof of Lemma 2. The proof is done by induction on the knowledge order $j \in \mathbb{N}$.

- Base case, $j = 0$:
  $F_0''(h) = last(h)$ so $\Cap F_0''(h) = last(h)$
- Induction step:
  Induction hypothesis:

$$\forall h \in H_G,\ \Cap F_j''(h) = last(h)$$

  Proof for $j + 1$:

$$\begin{aligned}
\Cap F_{j+1}''(h) &= \Cap \left(\{F_j''(h') \mid h' \sim_i h\}\right)_{i\in\mathcal{A}} && \textit{(Def. 2.5.1)}\\
&= \bigcap_{i\in A}\{\Cap F_j''(h') \mid h' \sim_i h\} && \textit{(Def of } \Cap\textit{)}\\
&= \bigcap_{i\in A}\{last(h') | h' \sim_i h\} && \textit{(induc. hyp.)}\\
&= last(h) && \textit{(PDK)}
\end{aligned}$$

□

Now that Lemma 2 is proven, let us prove Theorem 2

Proof of Theorem 2. The proof is done by structural induction on $\phi \in \mathcal{L}_n^{jK}$.

- Base case:

$$\begin{aligned}
&\mathcal{M}_{H_G}, h \models p\\
&\iff p \in L_{H_G}(h) && \textit{(Semantics)}\\
&\iff p \in \{p_{last(h)}\} && \textit{(Labelling function for } \mathcal{M}_{H_G}\textit{)}\\
&\iff p = p_{\Cap F_j''(h)} && \textit{(Lemma 2)}\\
&\iff F_j''(h) \models_B p && \textit{(Semantics)}
\end{aligned}$$

- Induction step:
  The induction step for $\neg\phi$ and $\vee$ are direct. Refer to proof of Theorem 1 for more details. Let us prove the induction step for the K_i operator. Assume ϕ of alternation depth at most $j - 1$ and:

$$\mathcal{M}_{H_G}, h \models \phi \iff F_{j-1}''(h) \models_B \phi$$

We have $K_i\phi$ of alternation depth at most j and:

$$
\begin{aligned}
& F''_j(h) \models_B K_i\phi \\
& \iff F''_j(h)[i] \models_A \phi && \textit{(Semantics)} \\
& \iff \forall b \in F''_j(h)[i], b \models_B \phi && \textit{(Semantics)} \\
& \iff \forall b \in \{F''_{j-1}(h') \mid h \sim_i h'\}, b \models_B \phi && \textit{(Def. 2.5.1)} \\
& \iff \forall h' \in H_G, h \sim_i h' \implies F''_{j-1}(h) \models_B \phi \\
& \iff \forall h' \in H_G, h \sim_i h' \implies \mathcal{M}_{H_G}, h' \models \phi && \textit{(Induc. hypo.)} \\
& \iff \mathcal{M}_{H_G}, h \models K_i\phi && \textit{(Semantics)}
\end{aligned}
$$

□

With this representation, any formula with alternation depth bounded by the knowledge order can be evaluated. Moreover, contrary to game-based representation, here each agent only needs its own local knowledge structure and not the global one. When evaluating a formula of the shape $K_i\phi$, agent i looks into its own local knowledge structure and verifies whether ϕ is true in every global knowledge structure it considers possible.

Example 17. Let us consider an example derived from Example 11 to illustrate Theorem 2 and its limitations. To simplify the example, we will only look at a second-order knowledge structure, hence the formula we will be using is a subformula of the one used in Example 11:
Let $h = start\ (grab, grab)\ bad$. Then the second order global knowledge is:

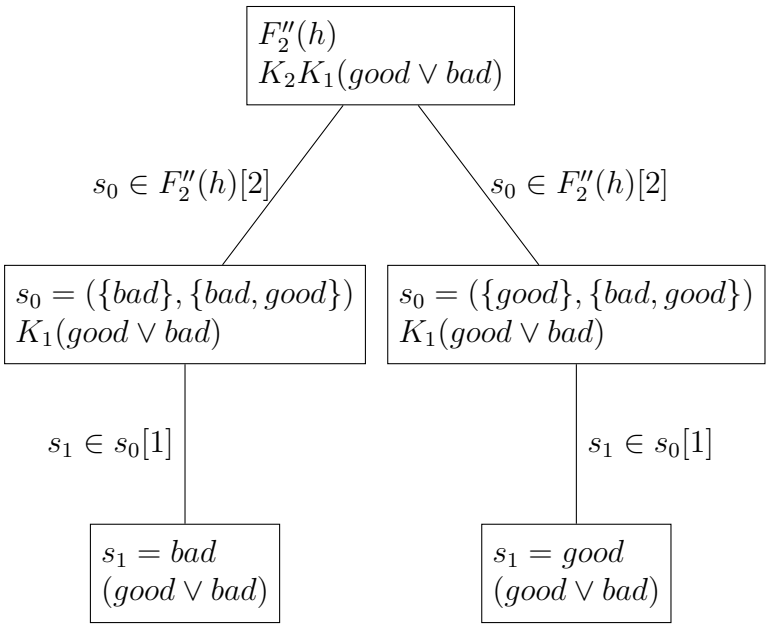
$$
\begin{aligned}
F''_2(h) = \Big(\big\{ & \ (\{bad\}, \{bad, good\}) \big\} \\
& \big\{ \ (\{bad\}, \{bad, good\}), \\
& \quad (\{good\}, \{bad, good\}) \ \big\} \Big)
\end{aligned}
$$

Let $\phi = K_2K_1\,(bad \vee good)$. ϕ is of alternation depth 2 since it can be rewritten $\phi = K_2K_1 bad \vee K_2K_1 good$ with both sub-formula with alternation depth 2. According to Theorem 1 and Theorem 2, we have:

$$
\begin{aligned}
& \mathcal{M}_{F''_2}, F''_2(h) \models K_2K_1\,(bad \vee good) \\
& \iff \mathcal{M}_{H_G}, h \models K_2K_1\,(bad \vee good)
\end{aligned}
$$

And as shown in Example 11, the formula is evaluated to true when using

the history-based interpretation. Hence, the right member of the above equivalence is true, and so is the left member. We can manually verify this. Here is the corresponding evaluation tree:



This evaluation tree is the same as the one obtained when evaluating the formula using the knowledge structure-based approach from Example 16. Only $\hat{\delta}^2(h)$ was replaced by $F_2''(h)$ which are equal according to Lemma 1. Finally, $bad \in \{good, bad\}$ so the left leaf is verified, and $good \in \{good, bad\}$ so the right leaf is also verified, and the formula evaluates to $true$.

We now have a representation which can be used in a practical setting without communication between agents. However, this representation can only be used to evaluate formula with bounded alternation depth. Let us try to look at a different approach, relying on game expansions instead of directly relying on knowledge structures.

4.2 Game-Based Interpretation

The second alternative Kripke structure is a game-based interpretation, introduced by Gurov et al. in [3]. The idea is to use the expanded game,

and use its set of locations as the states of the Kripke structure, and the indistinguishability relations as accessibility relations.

Definition 4.2.1 (Game-Based Kripke Structure)**.**
Let G be a MAGIIAN for which the PDK condition is respected. For any $j > 0$, its j-expansion $G^{jK} = (S^{jK}, s_I, \Sigma, \Delta^{jK}, \mathcal{O}^{jK})$ induces a Kripke structure, $\mathcal{M}_{G^{jK}} = \{S^{jK}, \{\sim_i\}_{i \in Agt}, L\}$ defined as follows:

- $\forall s, s' \in S^{jK}, \quad s \sim_i s' \iff s[i] = s'[i]$, with $s[i]$ the i-th component of s.
- $\forall s \in S^{jK}, \quad L(s) = \{p_{\mathbb{m}s}\}$

Gurov et al. cautiously conjecture that the Kripke structure induced by the j-the expansion of a game can be used to evaluate any formula of epistemic logic with an alternation depth at most j.

Conjecture 1. *Let G be a MAGIIAN, and $\mathcal{M}_{H_G}$ be the history-based Kripke structure induced by H_G as defined in Def. 2.2.9. For any $j > 0$, let G^{jK} be the j-th expansion of G, and $\mathcal{M}_{G^{jK}}$ be the game-based Kripke structure induced by G^{jK}. For any $h \in H_G$, and $\phi \in \mathcal{L}_n^{jK}$, we have:*

$$\mathcal{M}_{H_G}, h \models \phi \iff \mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models \phi$$

In this report, this conjecture will not be proven. However, and even if this does not come near a valid mathematical proof, brute-force testing was made to find counterexamples within the cup-lifting game. To do so, the truth value of the evaluation of each formula with alternation depth at most 4 in each history of length at most 5 has been compared to the same evaluation using the history-based model. No example was found in which the evaluation of the formula yielded different results for the history-based interpretation and the game-based interpretation. Note that the game stabilises after 2 expansions [1]. Therefore, testing formulas with alternation depth greater than 4 should not reveal any new counterexample. Additionally, the game loops on the joint action *(squeeze, squeeze)* in the state *good*, and also loops on both *lose* and *win* states. Each iteration of any of these loops preserves the knowledge structures, and does not change the truth value of a formula. Therefore, a history looping on one (or more) of these states can be replaced with a corresponding history where each looping action and resulting state has been removed, and the evaluation of the formulas for this new history will get the same results. Thus, exploring histories of length greater than 5 should not

provide a new counterexample.
This representation cannot directly be used to evaluate any formula of epistemic logic with unbounded alternation depth. Let us look at an example from the cup-lifting game to illustrate this limitation.

Example 18. Let $h = start\ (grab, grab)\ bad\ (squeeze, squeeze)\ good$, and $\phi = K_1 K_2 good$. The alternation depth of ϕ is 2. Let us try to evaluate the formula by using only the first game-expansion.

The first order global knowledge structure is: $\hat{\delta}^1(h) = (\{good\}, \{good\})$. During the first evaluation step, agent 1 will retrieve all global knowledge structures in S^{1K} with a first coordinate equal to $\{good\}$. In particular, this will also retrieve $s = (\{good\}, \{good, bad\})$ which is the global knowledge structure associated with history $h' = start\ (grab, grab)\ good$. Then we can see directly that $K_2 good$ does not hold at s, even if it holds at $\hat{\delta}^1(h)$. Thus, the formula evaluates to $false$. However, the evaluation of the formula through the history-based representation is $true$.

One way to see what happened is to look at the graphs of the base game and the expanded game. In the base game, we only have two states that are indistinguishable (Figure 2.1). After the first expansions, another couple of states are indistinguishable (Figure 2.2). This new couple of indistinguishable state is exactly s and $\hat{\delta}^1(h)$ from our example.

Note that in this particular case, one could think that the issue is that s and $\hat{\delta}^1(h)$ are two global knowledge structures resulting from histories of different length. Actually, for the cup-lifting game, adding the length of the history as a criterion for the accessibility relation solves the issue. It makes the states s and $\hat{\delta}^1(h)$ distinguishable by agent 1. However, this is not a reliable fix for the general issue. To see the limitations of this fix, we can modify the cup-lifting game. We add an action $wait$. When both agents $wait$, the state of the game remain the same. If exactly one agent uses $wait$, then the state of the game goes to $lose$. With this modified game, the histories h and h" $= start\ (grab, grad)\ good\ (wait, wait)\ good$ have the same length and we have $\hat{\delta}^1(h") = s$, which exactly replicates the setting of the faulty evaluation from before.

From Conjecture 1, Theorem 1, and Theorem 2, we naturally get the following corollary conjecture.

Corollary 1. *Let G be a MAGIIAN and $\mathcal{M}_{H_G}$ be the history-based Kripke structure induced by H_G. Let $h \in H_G$ be a history of G, $\phi \in \mathcal{L}_n^{jK}$ be a formula of alternation depth at most j:*

$$\begin{aligned} & \mathcal{M}_{H_G}, h \models \phi \\ \iff & \mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models \phi \\ \iff & \mathcal{M}_{\hat{\delta}^j(h)}, \hat{\delta}^j(h) \models \phi \\ \iff & F''_j(h) \models_B \phi \end{aligned}$$

Assuming Conjecture 1 holds, the proof is direct, by transitivity and symmetry.

Although a proof of Conjecture 1 will not be given, the following lemma might still be of use for a complete formal proof.

Lemma 3. *Let G be a game. For any two histories $h, h' \in H_G$, any agent i, and any knowledge order j, we have:*

$$h \sim_i h' \implies \hat{\delta}^j(h) \sim_i \hat{\delta}^j(h')$$

where $\hat{\delta}^j(h) \sim_i \hat{\delta}^j(h') \iff \hat{\delta}^j_i(h) = \hat{\delta}^j_i(h')$.

This lemma is from a previous work on MAGIIAN by Carl Åkerman Rydbeck and the full proof can be found in [6]. Note that the converse of the lemma does not hold in general.

Example 19. Consider the following histories from the cup-lifting game:

$$\begin{aligned} h_1 &= start\ (grab, grab)\ good \\ h_2 &= start\ (grab, grab)\ bad \end{aligned}$$

We have $h_1 \sim_2 h_2$ and:

$$\hat{\delta}^1(h_1) = (\{good\}, \{good, bad\}) \sim_2 (\{bad\}, \{good, bad\}) = \hat{\delta}^1(h_2)$$

Now if we play one more step and get the following histories:

$$\begin{aligned} h_1 &= start\ (grab, grab)\ good\ (squeeze, squeeze)\ good \\ h_2 &= start\ (grab, grab)\ bad\ (squeeze, squeeze)\ good \end{aligned}$$

Then we also have $\hat{\delta}^1(h_1) \sim_1 \hat{\delta}^1(h_2)$ since $good \sim_1 good$, however $h_1 \not\sim_1 h_2$,

since agent 1 can distinguish between the $good$ and bad states after the joint $(grab, grab)$ action.

Additionally, the equivalence $\mathcal{M}_{H_G}, h \models p \iff \mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models p$, for any $j > 0$ and any h, and where $p \in \mathcal{AP}$ is an atomic proposition, is a direct consequence of Lemma 2.

To better understand how the game-based interpretation from Conjecture 1 differs from the interpretation based on knowledge structures, let us look at an example from the cup-lifting game.

Example 20 (Example of Conjecture 1).
Let us consider the same setting as Example 11.
Let G be the cup-lifting game from Figure 2.1, and G^{2K} be its second expansion. Finally, let $h = start\ (grab, grab)\ bad$, and $\phi = K_2 K_1\ (good \vee bad)$.

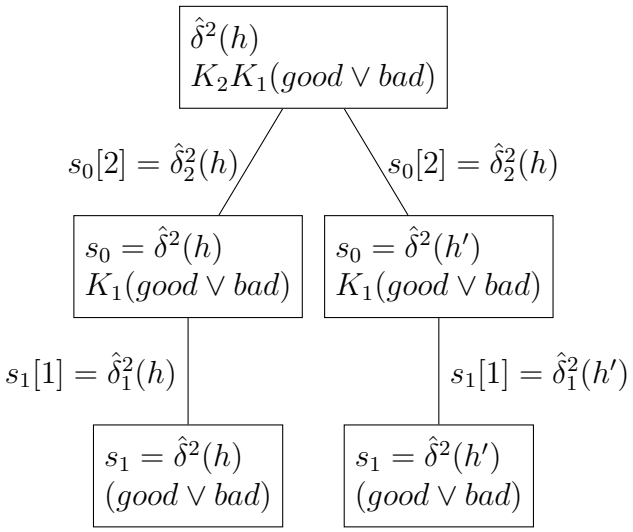
This formula is evaluated to $true$ at h when the history-based approach is used, as shown in Example 11, and confirmed in Example 17. Thus, if Conjecture 1 holds, we should also have $\mathcal{M}_{G^{2K}}, \hat{\delta}^2(h) \models \phi$. Let us verify it manually. Before anything, we need to compute the global knowledge $\hat{\delta}^2(h)$:

$$\hat{\delta}^2(h) = F_2''(h) = \Big(\big\{\ (\{bad\}, \{bad, good\})\ \big\} \\ \big\{\ (\{bad\}, \{bad, good\}), \\ (\{good\}, \{bad, good\})\ \big\}\Big)$$

We can also define $h' = start\ (grab, grab)\ good$, and compute $\hat{\delta}^2(h')$. This will be helpful for the first evaluation step.

$$\hat{\delta}^2(h') = \Big(\big\{\ (\{good\}, \{bad, good\})\ \big\} \\ \big\{\ (\{bad\}, \{bad, good\}), \\ (\{good\}, \{bad, good\})\ \big\}\Big)$$

Then, we can construct the evaluation tree:



Finally, for both leaves the intersection collapses to a singleton (either $good$ or bad). Thus, both leaves are verified, and the formula evaluates to true, which is the expected result.

During the evaluation of the formula in the previous example, each time a knowledge operator K_i is encountered, the i-th coordinate of a global knowledge structure is accessed, and every knowledge structure that has the same value as its i-th coordinate is accessed. Since the i-th coordinate is the set of states considered possible by agent i, this means we are exploring all worlds (i.e. histories) that agent i considers possible.

This game-based interpretation can also be used in a setting where agents do not communicate. Let $\phi = K_i\psi \in \mathcal{L}_n^{jK}$ and $h \in H_G$. Since the formula starts with K_i, agent i should be able to evaluate it locally using a game-based Kripke structure. Agent i has a copy of S^{jK}, as well as a copy of its own local knowledge structure $\hat{\delta}_i^j(h)$ for some knowledge order greater or equal to the alternation depth of the formula. The first step of the evaluation of ϕ is to retrieve all elements of S^{jK} such that their i-th coordinate is equal to $\hat{\delta}_i^j(h)$. To do so, only the local copy of S^{jK} and the local copy of $\hat{\delta}_i^j(h)$ are used. Hence, it can be done locally. Then, for each element of S^{jK} previously retrieved, agent i needs to evaluate ϕ. Since agent i has a full copy of S^{jK}, it can access any coordinate of the elements of S^{jK} that it retrieved in the first evaluation step. That is to say, for any global structure retrieved in the first evaluation step, agent i can access not only the i-th coordinate, but also any other coordinate.

There is, however, one limitation to the local evaluation of a formula. For

a formula to be evaluated locally by agent i, the formula must be of the shape $\phi = K_i\psi$, or $\phi = \bigvee_{p \in P \subseteq \mathcal{AP}} p$. For example, the formula $\phi = K_i\psi \vee K_{i'}\Psi$ cannot be evaluated locally, neither by agent i, nor by agent i'. It would require access to the local knowledge structures of agent i and agent i' which cannot be done locally without communication between agents. Thus, each even if each disjunct can be evaluated separately by different agents, the complete formula cannot. In a practical setting, this limitation might not be an issue, as asking agent i to evaluate a formula about agent i' knowledge often does not make sense.

Example 21.
Let G be the cup-lifting game, $\phi = (K_2(bad \vee good)) \vee (K_1(bad))$, and $h = start\,(grab,\ grab)\,bad$. Using the history-based interpretation, this formula evaluates to true. Using the game-based interpretation provides the same result. However, the result cannot be obtained through local evaluation.

$$\mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models (K_2(bad \vee good)) \vee (K_1(bad))$$
$$\iff \Big(\mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models K_2(bad \vee good)\Big) \vee \Big(\mathcal{M}_{G^{jK}}, \hat{\delta}^j(h) \models (K_1 bad)\Big)$$

The first disjunct can be evaluated locally only by agent 2, and the second disjunct can be evaluated locally only by agent 1. Practically speaking, asking agent 2 to evaluate locally the whole formula means asking agent 2 if the sensor of agent 1 detected a bad grip. This does not make sense if agents cannot communicate. It could make sense to ask if agent 2 *knows* if agent 1 detected a bad grip, but this gives a completely new formula $\phi' = (K_2(bad \vee good)) \vee (K_2K_1(bad))$ which can be evaluated locally by agent 2, and evaluates to false (be it using history-based or game-based interpretation).

There is one particular setting in which certain formulas of the shape $\bigvee_{i \in A \subseteq Agt} K_i\psi_i$ can be evaluated locally. To highlight this case, a closer examination of j-stable games is required.

4.3 Local Interpretation Based on Stable Games

4.3.1 Characterization of Stable Games

As mentioned in Def. 2.2.8, expanded games result from the iterated application of the Multi-agent Knowledge-Based Subset Construction (MKBSC). In this process, MAGIIAN might *stabilise* with respect to the action and observation relations, as highlighted by Gurov et al. [1]. Intuitively, increasing the knowledge order of agents in a stabilised game does not improve the ability to distinguish between states. All relevant information of higher order knowledge is already contained in a lesser order knowledge.

Definition 4.3.1 (j-stable Game)**.**
Let G be a MAGIIAN. G is termed *j-stable* if G^{jK} is isomorphic to $G^{(j+1)K}$, with respect to both the action and the observation relations, and j is the smallest natural fitting this criterion. A game G is termed *stable* if it is 0-*stable*.

The isomorphism preserves the action and observation relations. They are respectively visually represented by the edges and dashed-edges in the visual representation of the game graphs. Therefore, on can intuitively view j-stable games as games where the graphs of the $j-th$ and $(j+1)-th$ expansions are the same if we ignore the labels of each node.
In order to study the specific case of j-stable games, we aim to exhibit such an isomorphism.

Definition 4.3.2.
Recall the definition of S^{jK} for expanded games from Def. 2.2.8.

We define the following mappings:

$$\begin{aligned} \chi : S^{jK} &\longrightarrow S^{(j+1)K} \\ s &\longmapsto \Big(\big\{ s' \in S^{jK} \mid s[i] = s'[i] \big\} \Big)_{i \in Agt} \end{aligned}$$

$$\begin{aligned} \xi : S^{(j+1)K} &\longrightarrow S^{jK} \\ s &\longmapsto \cap s \end{aligned}$$

Where $\cap s$ is an abuse of notation defined by $\{\cap s\} = \bigcap_{i \in Agt} s[i]$. This abuse of notation will be justified below, in the proof of Theorem 3.

This definition is a generalised definition of the isomorphism from Loc to S^{1K} introduced in [5]. The generalisation is made by Gurov et al. in [3], where G is conjectured to be j-stable if and only if χ is an isomorphism. We will now prove this claim.

Theorem 3. *Let G be a MAGIIAN. G is j-stable if and only if χ is an isomorphism between S^{jK} and $S^{(j+1)K}$, with respect to both the action and observation relations, and j is the smallest such number. Furthermore, if G is j-stable, ξ is the inverse of χ.*

Proof of Theorem 3.

The theorem consists of an equivalence between G being j-stable, and χ being an isomorphism. First, the by definition of a j-stable game, if χ is an isomorphism with respect to both the action and observation relations, then G is j-stable.

Only the other implication remains to be proven: if G is j-stable, then χ is an isomorphism. The proof of this implication is divided in two steps. First, we prove that if G is j-stable, then χ is bijective (with ξ its inverse). Then, we prove that if G is j-stable, then χ is structure preserving (with respect to the action and observation relations).

- Assume G is j-stable. We will prove that χ is bijective. By definition of ξ and χ, for any $s \in S^{jK}$ we have:

$$\begin{aligned}\xi(\chi(s)) &= \cap\chi(s) && \textit{(Def. 4.3.2)}\\ &= \cap((\{s' \in S^{jK} \mid s'[i] = s[i]\})_{i\in\mathcal{A}}) && \textit{(Def. 4.3.2)}\end{aligned}$$

and

$$\begin{aligned}&\{\cap((\{s' \in S^{jK} \mid s'[i] = s[i]\})_{i\in\mathcal{A}})\}\\ =&\bigcap_{i\in\mathcal{A}}\{s' \in S^{jK} \mid s[i] = s'[i]\} && \textit{(Notation } \cap\textit{, see Def. 4.3.2)}\\ =&\{s' \in S^{jK} \mid \forall i \in \mathcal{A}, s'[i] = s[i]\} && \textit{(Set theory)}\\ =&\{s\}\end{aligned}$$

hence:

$$\forall s \in S^{jK}, \xi(\chi(s)) = s$$

And χ is bijective with ξ for inverse. Note that $\bigcap_{i\in Agt}\chi(s)[i] = \{s\}$ is a singleton since the resulting set is the set of vectors which have the exact

same coordinate as s, i.e. only s itself. This is the reason for the abuse of notation $\cap s$, since ξ is only ever applied to some $\chi(s)$. The bijectivity of χ is therefore proven and only the structure preserving criterion remains to be verified.

- Assume G is j-stable. We will prove that χ is structure preserving with respect to the action and observation relations.

  - χ preserves the observation relation:
    We want to prove:

    $$\forall s_1, s_2 \in S^{jK}, \forall i \in \mathcal{A} : \\ s_1 \in O_i^{jK}(s_2) \iff \chi(s_1) \in O_i^{(j+1)K}(\chi(s_2))$$

    We have:

    $$\begin{aligned} &\chi(s_1) \in O_i^{(j+1)K}(\chi(s_2)) \\ &\iff \chi(s_1)[i] = \chi(s_2)[i] && \textit{(Def. 2.2.8)} \\ &\iff \{s' \in S^{jK} \mid s_1[i] = s'[i]\} \\ &\qquad = \{s' \in S^{jK} \mid s_2[i] = s'[i]\} && \textit{(Def. 4.3.2)} \\ &\iff s_1[i] = s_2[i] \\ &\iff s_1 \in O_i^{jK}(s_2) && \textit{(Def. 2.2.8)} \end{aligned}$$

    Thus, the observation relation is proven.

  - χ preserves the action relation:
    Let be $s_1, s_2 \in S^{jK}$, we want to prove:

    $$(s_1, \sigma, s_2) \in \Delta^{jK} \iff (\chi(s_1), \sigma, \chi(s_2)) \in \Delta^{(j+1)K}$$

    Let us first show:

    $$(\chi(s_1), \sigma, \chi(s_2)) \in \Delta^{(j+1)K} \implies (s_1, \sigma, s_2) \in \Delta^{jK}$$

    We will prove its contrapositive. Assume $(s_1, \sigma, s_2) \notin \Delta^{jK}$. Then, by definition of χ and according to the *composition* step described in [1], the transition $(\chi(s_1), \sigma, \chi(s_2))$ will be pruned. Hence, $(s_1, \sigma, s_2) \notin \Delta^{jK} \implies (\chi(s_1), \sigma, \chi(s_2)) \notin \Delta^{(j+1)K}$. Thus, the contrapositive holds and the one way implication is proven.
    Let us now prove:
    $(s_1, \sigma, s_2) \in \Delta^{jK} \implies (\chi(s_1), \sigma, \chi(s_2)) \in \Delta^{(j+1)K}$. This time,

we proceed by contradiction. Assume there exists $s_1, s_2 \in S^{jK}$ such that $(s_1, \sigma, s_2) \in \Delta^{jK}$ but $(\chi(s_1), \sigma, \chi(s_2)) \notin \Delta^{(j+1)K}$.
From the previous result, we know that for each transition $(\chi(s'), \sigma, \chi(s'')) \in \Delta^{(j+1)K}$ there is a transition $(s', \sigma, s'') \in \Delta^{jK}$. And since $(s_1, \sigma, s_2) \in \Delta^{jK}$ but $(\chi(s_1), \sigma, \chi(s_2)) \notin \Delta^{(j+1)K}$, we have $card\left(\Delta^{jK}\right) > card\left(\Delta^{(j+1)K}\right)$, where $card(A)$ denotes the cardinality of set A. However, since G is j-stable, we also have $card\left(\Delta^{jK}\right) = card\left(\Delta^{(j+1)K}\right)$, which is a contradiction with the previous result. Hence, the assumption is wrong, and the implication is proven.

Both one-way implications are proven, therefore χ preserves the action relation.

Under the assumption that G is j-stable, we proved χ to be a bijection preserving the action and observation relations. Hence, if G is j-stable, then χ is an isomorphism. Therefore, both one-way implications are proven and so is the theorem. □

Now that j-stable games are properly characterized, local formula evaluation can be studied.

4.3.2 Local Evaluation of Epistemic Logic Formulas

j-stable games are MAGIIAN, hence Conjecture 1 provides a way to evaluate formula locally. However, as mentioned previously, local evaluation through Conjecture 1 cannot be done with formula of the shape $\phi = \bigvee_{i \in A \subseteq Agt} K_i \psi_i$, with $\psi_i \in \mathcal{L}_n$ for any $i \in Agt$.

Lemma 4 (Local Formula Evaluation in Stable Games)**.**
Assume Conjecture 1 holds. Let G be a j-stable game. Let $\phi = \bigvee_{i \in A \subseteq Agt} K_i \psi_i$ with $\psi_i \in \mathcal{L}_n$ for any i. If there exists $k \in Agt$ such that:

$$\forall i \in A \setminus \{k\}, alt_depth(K_i \psi_i) \geq j$$

then agent k can evaluate the formula locally using the game-based interpretation by adding K_k operator to the left of the formula.

Proof. Assume the premise holds. Since the game is j-stable, adding a knowledge operator at the left of a formula of alternation depth greater or equal to j does not change the truth value of the formula. Thus, distributing the K_k operator to each disjunct results in one of the following case for each disjunct:

- $i \neq k$, then $alt_depth(K_k K_i \psi_i) \geq j+1$ and $K_k K_i \psi_i$ evaluates to the same as $K_i \psi_i$ since the game is j-stable.
- $i = k$, then distributing the K_k operator we have: $K_k K_k \psi_k$ which collapses to $K_k \psi_k$ by semantics.

Thus, the formula can be evaluated locally by agent k after prepending a K_k operator to the formula. □

Remark. Instead of prepending a K_k operator to the formula, one might keep the initial formula but tweak the first evaluation step. Let G be a j-stable MAGIIAN, $\phi = \bigvee_{i \in A \subseteq Agt} K_i \psi_i$, and $k \in Agt$ such that:

$$\forall i \in A \setminus \{k\}, alt_depth(K_i \psi_i) \geq j$$

Then let $h \in H_G$ and let $j' > 0$ be a knowledge order. For agent k to evaluate the formula locally at point $\hat{\delta}^{j'}(h)$, for each disjunct the first step of the game-based approach can be replaced by:

$$\begin{aligned}
&\mathcal{M}_{G^{j'K}}, \hat{\delta}^{j'}(h) \models K_i \psi_i \\
&\iff \forall s \in S^{jK}, \left(\exists s_k \in \chi(\hat{\delta}^{j'}(h))[k], s_k[i] = s[i]\right) \implies \mathcal{M}_{G^{j'K}}, s \models \psi_i
\end{aligned}$$

where χ is the isomorphism defined in Def. 4.3.2.
This is mathematically equivalent to prepending a K_k operator. For any $i \in A$:

$$\begin{aligned}
&\mathcal{M}_{G^{j'K}}, \hat{\delta}^{j'}(h) \models K_k K_i \psi_i \\
&\iff \forall s_k \in S^{jK}, s_k[k] = \hat{\delta}^{j'}_k(h) \implies \mathcal{M}_{G^{j'K}}, s_k \models K_i \psi_i \\
&\iff \forall s_k \in S^{jK}, \\
&\qquad s_k[k] = \hat{\delta}^{j'}_k(h) \implies \left(\forall s \in S^{jK}, s[i] = s_k[i] \implies \mathcal{M}_{G^{j'K}}, s \models \psi_i\right) \\
&\iff \forall s \in S^{jK}, \\
&\qquad \left(\exists s_k \in S^{jK}, s_k[k] = \hat{\delta}^{j'}_k(h) \wedge s_k[i] = s[i]\right) \implies \mathcal{M}_{G^{j'K}}, s \models \psi_i \\
&\iff \forall s \in S^{jK}, \left(\exists s_k \in \chi(\hat{\delta}^{j'}(h)[k], s_k[i] = s[i]\right) \implies \mathcal{M}_{G^{j'K}}, s \models \psi_i
\end{aligned}$$

Example 22 (Local Evaluation in Stable Games).
The cup-lifting game is 2-stable [1]. Previously, in Example 17, we evaluated $K_2 K_1(bad \vee good)$ at point $h = start(grab, grab)bad$. The formula evaluated to $true$, and was evaluated locally by agent 2. Since the game is 2-stable, and

the formula has an alternation depth of 2, it can also be evaluated locally by agent 1. According to Lemma 4, this evaluation can be done by prepending K_1 to the formula. This results in $K_1K_2K_1(bad \vee good)$, which is the formula evaluated locally by agent 1 at the same point h in Example 20.

Chapter 5

Conclusions and Future work

5.1 Conclusions

5.1.1 General Conclusions

We managed to prove that an interpretation based on knowledge structures can be used to evaluate any formula of *epistemic logic* with an alternation depth bounded by the knowledge order. Moreover, this evaluation can be done locally by each agent, without requiring any form of communication. Therefore, this models does better than the history-based model both in term of local evaluation —impossible with the history-based interpretation that requires each agent to know the actions taken by other —, and in term of spatial complexity for games with ”long” histories. Compared to the k-trees introduced by van der Meyden [2], given a game history, the knowledge structure contains exactly the leaves of the corresponding *k-trees*. Hence, this model could provide a *lighter* alternative in terms of spatial complexity.

We conjectured a game-based approach can be used for formulas with bounded alternation depth. A brute-force test was carried out to try to find counterexample in the cup-lifting game, but none was found. This approach also allows for a local evaluation by agent k of any formula that can be written as $K_k\phi$, with ϕ a formula of *epistemic logic* with bounded alternation depth.

Finally, we studied the case of stable games. We characterized j-stable games using a function that is an isomorphism if and only if the MAGIIAN considered is j-stable. In such a case, the condition for local evaluation is different, permitting more formulas to be evaluated locally.

5.1.2 Practical Conclusions

In the context of knowledge-based programs, we must differentiate between formulas with and without a bound on the alternation depth. In the context of bounded alternation depth, then each agent can compute its local knowledge structure with a knowledge-order equal to the alternation depth, and update it throughout the play. Each formula can then be evaluated directly on its local knowledge structure. This computation might be more efficient than relying on the game-based representation, since it only requires recursive accesses to coordinates of its local knowledge structure.

In most practical settings, the alternation depth of the epistemic logic formulas used in a knowledge-based program can be bounded. As long as the code of the program is fixed and completely written before the start of the execution, one can always compute the alternation depth of each formula, and then use the maximum alternation depth as the bound (and as the knowledge-order of the model). If the code of the program itself were to evolve without restrictions during the execution, then the alternation depth could be unbounded. In such a hypothetical setting, there would be a need for another approach that supports unbounded alternation depth.

5.2 Future work

The first thing to be done is to study Conjecture 1 on the completeness and soundness of the game-based interpretation. Then, if the conjecture indeed holds, the complexity should be studied, and practical applications investigated.

Further work also remains to be done on the interpretation based on knowledge structures. As mentioned previously, this approach could provide an alternative to *k-trees* with a lower space complexity. This should be further studied, along with time complexity.

Finally, investigating to find a model that can handle formulas with unbounded alternation depth could also be done. This would allow for more expressive power in games that do not stabilize. It would also most-likely allow for an extension of the syntax and semantics to include fixed point recursion (*epistemic μ-calculus*).

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